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Research Statement

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When a developer is tasked with creating a software application, accessibility is rarely considered as a first-class requirement, despite how an inaccessible application can be difficult, or downright impossible to use for the 16% of the global population who experience some form of disability [11]. Primarily attributed to a lack of organizational prioritization and pedagogical awareness, software inaccessibility has been addressed with engineering techniques that automatically detect inaccessible components in web, mobile and desktop applications. Built on rules and heuristics, such techniques are easily scalable, and effective in passing compliance checks. *But how well do they represent what disabled people actually experience?*

My research addresses this question and builds LLM-based approaches that put disabled people's experience at the center of accessibility testing, using AI to scale their insight rather than replace it. Sitting between software engineering and human-computer interaction, my work combines empirical studies of practice, user studies with disabled testers, and evaluations of the LLM coding agents that increasingly write the software people use. It has produced three first-author papers at premier human-computer interaction and software engineering venues (CHI and ICSE respectively), including an ICSE 2025 Distinguished Paper, and further collaborative papers at ICSE, FSE, ASE, and ASSETS.

Thread 1: How accessibility materializes in practice

To understand why the software development ecosystem produces inaccessible products, interview and survey studies have been conducted with software professionals [12], [14], [15]. Common obstacles included lack of awareness about accessibility concepts and best practices, along with a lack of incentive to put in the time and effort to learn and execute them. In "#A11ydev" (CHI '23) [10], I aimed to further our understanding of the space by looking into social media, Twitter specifically, where conversations on accessibility converged user and developer silos. Across three viewpoints of process, product and people, we found obstacles ranging from the individual to the organization, in development practices and institutional cultures. Conversely, an increasing advocacy for and training on accessibility in software companies, paired with the development of accessibility-focused design systems, frameworks and testing tools, also shone a light on better roads ahead.

From this holistic view of software practices, two specific findings influenced my research going forward:

- *User testing is more insightful than automated tools.* Automatically auditing software had become the industry standard, despite how inaccessible the final product remained. While user testing was advocated for its high fidelity and nuanced insights, it was deemed too difficult and expensive for individual developers and even companies to recruit and collaborate with disabled testers.
- *People with disabilities voluntarily test products.* Upon receiving a new product or a new update, long threads of accessibility reports would be posted by disabled users. Or, if issues were found during use, the users would try to communicate with the developers directly, through social media, app stores or any other medium. We found an immediacy and enthusiasm in providing accessibility feedback from the users themselves.

Beyond social media mining, in a recent data-mining study of Moodle's issue repository (ASSETS '26, to be presented), my colleague and I examined how accessibility issues are reported and resolved in

an open source project. We report similar observations, such as a dependence on automated auditing, along with new insights on what resources are used to describe, recreate, test and fix issues. Most importantly, we observe how dedicated organizational efforts to improve accessibility have not only benefited the software's accessibility, but reduced time and effort in resolving issues as well.

Thread 2: Automated tools' alignment with disabled experiences

As part of the SEAL lab at UCI, I collaborated in developing highly effective accessibility testing techniques. My role in these collaborations focused on the human aspects, as I led the user studies with blind testers, who helped us recognize and characterize accessibility issues and evaluate the effectiveness of the developed tools. This included "over-accessibility" issues where Android app functionality became insecurely exposed through accessibility services (ASE '22) [13], hindrances caused by third-party advertisements embedded on Android app UI (ICSE '24) [9], locatability and actionability problems when navigating websites using screen readers (ASE '25) [8], and reflow-related issues in responsive web apps (FSE '26) [3]. The user studies were key in setting our research apart, as we addressed real issues and better understood the nuanced effects of inaccessibility, which further improved our techniques.

These studies also inspired me to take a more systematic look at similar accessibility detection techniques in research, and bridge their claims with real users' feedback. I conducted a mixed-methods study combining a systematic literature review and user studies with blind individuals (CHI '26) [4]. Upon synthesis, a shared language was developed to characterize and categorize screen reader issues on Android. A total of 28 issue types emerged across five categories: labeling, navigation, activation, dynamic change and feedback. It also revealed key trends and limitations of automated interventions.

- *Heuristics are scalable, but not faithful.* Automation techniques reliant on rules and heuristics can be deployed at scale to easily scan applications for accessibility, but they fail to cover the full range of issues faced by users. Issues often rely on semantic understanding and context.
- *An upwards trend towards fidelity.* Research has made strides from naive static checkers to dynamic testing, simulating assistive tech interactions on built apps, which increased issue coverage.
- *Testing alone cannot solve accessibility.* Echoing calls for shifting accessibility interventions earlier in the development lifecycle, as observed in #A11yDev, testers in our user studies suggested alternate interaction and navigation design which would not be addressed in the testing phase. Rather, design must consider assistive technologies and disability needs before implementation takes place.

From our categorization, we developed MCAG (Mobile Content Accessibility Guidelines), as a mobile-focused companion for WCAG, which serves as a foundational guideline for web accessibility. It provides a systematic reference to understand accessibility barriers and find software solutions that address them.

Thread 3: Streamlining user feedback as accessibility reports

Inspired by the need to make user testing more popular, and with access to LLMs as a technology capable of processing natural language, I built Reca11 (ICSE '25) [7]. In collaboration with Fable, a company connecting software companies with disabled testers, I analyzed their user testing process and retrieved their archival test sessions to pinpoint aspects in the process that can be automated, without removing the tester's feedback. Reca11 takes as input the transcript from a test session and, using LLMs

with engineered prompts and contexts, generates a test report for developers. The results demonstrated that Reca11 alleviated the effort to go through multiple hour long sessions, and the expertise required to parse accessibility feedback and connect to the software under test. Furthermore, the generated reports not only covered expert-annotated entries (98% recall, 85% precision), but unearthed new ones missed by the former. This work received a Distinguished Paper Award and the findings inspired Fable to deploy a similar test report summarization tool on their platform [5].

Continuing collaboration with Fable, I also led a study that compares the usability testing performance of testers with cognitive disabilities, against non-disabled testers. Currently under review [2], this work found that cognitive testers do not fall behind in conventional usability metrics, while providing more nuanced usability feedback. This study has also expanded my understanding of accessibility in the cognitive disabilities space.

My current work aims to expand on Reca11 to introduce user traces, runtime metadata and screenshots as input beyond spoken feedback. This data can be recorded while a screen reader user is testing an application, exposed via Android's accessibility services. The additional data points will reduce LLM-induced false positives, and detect barriers that testers cannot notice and therefore articulate, such as an element that cannot receive focus. This work brings together the insight from #A11yDev on voluntary testing, the foundation technique from Reca11, and the mobile accessibility taxonomy from MCAG, to build a system of independent accessibility testing for users with disabilities, and enable a more seamless communication channel to developers who can then solve real issues.

Thread 4: Are LLMs good at accessibility

Reca11 demonstrated that LLMs are capable of understanding accessibility implications of software products based on natural language feedback and guidelines as context. As LLM technologies have advanced to become autonomous coding agents, quickly democratizing the process of developing web applications, I asked whether this will lead to a more accessible internet or not. In a work currently under review [1], I conducted an empirical study with a set of three coding agents (Codex, Claude Code and OpenCode) and four models (GPT, Sonnet, DeepSeek and Qwen), engineering the harnesses for at-scale development and evaluating the accessibility of their generated websites.

The results were less than ideal, as all of the websites included some accessibility violations. Color contrast issues persisted across all 90 AI generated websites, with feature-specific issue prevalence, such as labeling mismatch in forms, heading and table issues in textual content, and more. The experiments then expanded to incorporate agentic skills dedicated to audit and fix accessibility. While still not perfect, a consistent improvement was observed, especially when employing specialized interventions, for instance, scripts to check color contrast. Analyzing agent trajectories also revealed a heightened reasoning about accessibility, influencing agent behavior. The results suggest a pathway towards accessible websites in the agentic era, on account of integrating purposive and high-fidelity techniques.

GenA11y (FSE '25) [6] was our first entry into LLM-powered accessibility issue detection. Addressing the biggest gap of existing checkers — lack of subjective and contextual understanding — GenA11y utilized the natural language and image processing capabilities of LLMs. While prior techniques checked, for instance, whether an alt text existed for an image, GenA11y can diagnose whether the existing alt text actually makes sense. GenA11y covers 37 WCAG success criteria, with future efforts to incorporate barriers caused by dynamically changing components.

Research agenda

1. Foreground disabled perspectives through software

My work on automating accessibility feedback from disabled testers has shown me the promise and importance of popularizing user testing as the next paradigm for software accessibility. The ultimate goal is to build a robust, scalable and accurate system, across boundaries of platform (web, mobile, desktop), disability (vision, motor, cognitive, etc.), and software artifact (product, code, design artifact).

The research core of this system will also evolve, from the current software engineering-focused design and implementation, to incorporating HCI methods and theories. For instance, participatory design, with disabled testers and developers, is necessary to build both sides of the system's interfaces: the tester-side recording tool, with attention to privacy, consent, and usability, and the developer-side interface, with a focus on creating actionable reports. The goal will shift from accuracy of the system towards its adaptability by the stakeholders.

2. Accessibility in the LLM era

LLMs now hold many roles in software engineering, each of which needs to be studied under the scope of accessibility. As *developers*, coding agents produce inaccessible websites. My empirical study on agents indicates that engineering harnesses with agentic interventions and analyzing their trajectories to find behavioral patterns are effective strategies to steer LLMs toward accessibility-aware development.

As *coding tools*, agents provide an ever evolving set of interfaces for programming: chatbots, IDE extensions, consoles and more. But in our accelerated churn, are we leaving behind disabled programmers, stretching again the disability gap? I will evaluate the accessibility implications of these interfaces, bringing in facets of trust, verifiability and creativity.

As *testers*, LLM-based detectors can judge systems based on semantic, contextual and multimodal input. Plugged to browser workers, LLMs can also simulate web navigation behavior and dynamically check for violations. But are they similarly vulnerable to the common pitfalls we found in our CHI '26 study [4]? I will investigate LLM-powered accessibility evaluators against human experiences and create ones that can complement tester feedback with scalable automation.

The common thread is that LLM's potential for enhancing accessibility in software use must be carefully validated and subsequently harnessed, while keeping disabled people's feedback as ground truth.

3. Human implications of software

Accessibility is my entry point into a broader question: whose experiences does software assume? My background in Bangladesh has made me acutely aware of how systems standing on the foundations of a different culture and language can further discriminate the already marginalized. With the rapid growth and adoption of expensive LLMs in our essential technologies, the promise of democratizing technological advancement in the era of AI may be misguided and ultimately counterproductive. The fundamentals of accessibility research has motivated and taught me the techniques of centering the human in developing and investigating software. I will inquire the quality of existing software systems and new ones built for and by LLMs under the dimensions of varying languages, cultural contexts, and economic and literacy levels.

In essence, I will pursue one question: *how can software be built and evaluated in ways grounded in the experiences of the people it serves?*

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